

Introduction to Probability and Applied Statistics

Week 5: Estimation and Large Samples

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Roadmap

Variance and covariance

Conditional expectation

Random samples and the sample mean

Sampling distributions and the law of large numbers

The central limit theorem

Estimators and their properties

Wrapping up

The computational identity

$$\begin{aligned}\mathbb{V}[X] &= \mathbb{E}[(X - \mu)^2] \\ &= \mathbb{E}[X^2 - 2\mu X + \mu^2] && \text{expand the square} \\ &= \mathbb{E}[X^2] - 2\mu \mathbb{E}[X] + \mu^2 && \text{linearity; } \mu \text{ is a constant} \\ &= \mathbb{E}[X^2] - 2\mu^2 + \mu^2 && \mathbb{E}[X] = \mu \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2\end{aligned}$$

“Variance equals the mean of the square minus the square of the mean.”

Covariance

In Week 2 we defined covariance in the sample:

$$\widehat{\text{Cov}}(X, Y) = \frac{1}{n-1} \sum_i (X_i - \bar{X})(Y_i - \bar{Y})$$

COVARIANCE

The **covariance** of X and Y in the population is

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] .$$

Properties of covariance

PROPERTIES

For constants a, b, c and d :

1. $\text{Cov}(X, X) = \mathbb{V}[X]$;
2. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$;
3. $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$;
4. $\text{Cov}(X + c, Y + d) = \text{Cov}(X, Y)$.

The variance of a sum

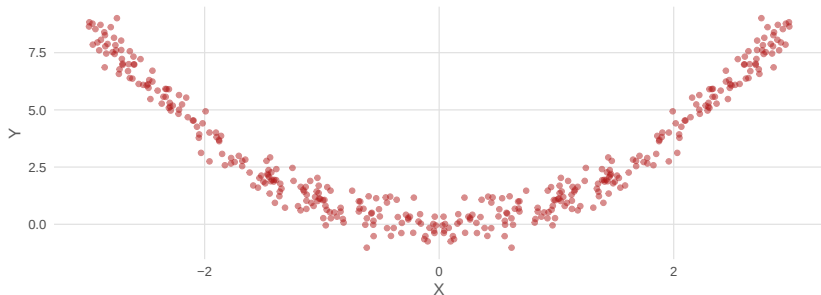
$$\mathbb{V}[X + Y] = \mathbb{V}[X] + \mathbb{V}[Y] + 2 \operatorname{Cov}(X, Y)$$

$$\mathbb{V}[X - Y] = \mathbb{V}[X] + \mathbb{V}[Y] - 2 \operatorname{Cov}(X, Y)$$

When X and Y are independent:

$$X \perp\!\!\!\perp Y \implies \operatorname{Cov}(X, Y) = 0 \implies \mathbb{V}[X \pm Y] = \mathbb{V}[X] + \mathbb{V}[Y]$$

Zero covariance is not independence



$Y = X^2 + \text{noise}$, and X is symmetric about zero, so $\text{Cov}(X, Y) = \mathbb{E}[X^3] = 0$.

$\text{Cov}(X, Y) = 0$ does not imply $X \perp\!\!\!\perp Y$. The implication runs in one direction only.

The operators

	Definition	Algebra
$\mathbb{E}[X]$	$\sum_i p_i x_i$	$\mathbb{E}[a + bX] = a + b\mathbb{E}[X]$ $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$, always
$\mathbb{V}[X]$	$\mathbb{E}[(X - \mu)^2]$	$\mathbb{V}[c] = 0$; $\mathbb{V}[X + c] = \mathbb{V}[X]$ $\mathbb{V}[cX] = c^2\mathbb{V}[X]$ $\mathbb{V}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$
$\text{Cov}(X, Y)$	$\mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$	$\text{Cov}(X, X) = \mathbb{V}[X]$ $\mathbb{V}[X + Y] = \mathbb{V}[X] + \mathbb{V}[Y] + 2\text{Cov}(X, Y)$

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Conditional expectation

CONDITIONAL EXPECTATION

The **conditional expectation** of Y given $X = x$ is the expected value of Y in the conditional distribution of Y given $X = x$:

$$\mathbb{E}[Y \mid X = x] = \sum_y y \mathbb{P}(Y = y \mid X = x)$$

The **conditional expectation function** assigns this number to each x :

$$\mathbb{E}[Y \mid X]: x \longmapsto \mathbb{E}[Y \mid X = x]$$

Evaluated at the random X , $\mathbb{E}[Y \mid X]$ is a random variable.

The law of iterated expectations

LAW OF ITERATED EXPECTATIONS

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]]$$

The outer expectation is taken over the distribution of X .

For a discrete X :

$$\mathbb{E}[Y] = \sum_x \mathbb{E}[Y \mid X = x] \mathbb{P}(X = x)$$

The inner expectation produces one number per value of x . The outer expectation averages those numbers, weighting each by the probability of the x that produced it.

Iterated expectations, term by term

$$\mathbb{E}[Y] = \sum_x \mathbb{E}[Y \mid X = x] \mathbb{P}(X = x)$$

x	$\mathbb{E}[Y \mid X = x]$	$\mathbb{P}(X = x)$	product
Not much	0.6521	0.0640	0.0418
Somewhat	0.6979	0.2975	0.2077
Very much	0.7132	0.6384	0.4553
sum			0.7047

```
mean(cc2$importance)
```

```
[1] 0.7047171
```

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Independent and identically distributed

IID

$X_1, \dots, X_n$ are **independent and identically distributed**, written $X_i \stackrel{\text{i.i.d.}}{\sim} F$, when every X_i has the same distribution F and any subcollection of them is mutually independent. Every X_i then has the same mean μ and the same variance σ^2 .

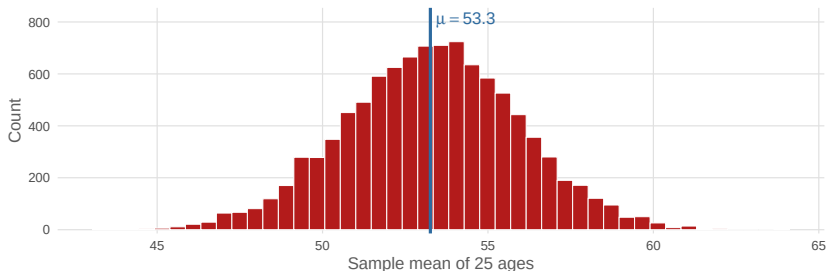
Identically distributed is a claim about the frame: every draw comes from the same population.

Independent is a claim about the procedure: one draw carries no information about any other.

The sample mean is a random variable

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Each X_i is a random variable, so $\bar{X}$ is a random variable. $\bar{X}$ has a distribution, an expected value and a variance of its own.



Ten thousand samples of $n = 25$.

The expected value of the sample mean

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n} \sum_i x_i\right]$$

The definition of $\bar{X}$.

$$= \frac{1}{n} \mathbb{E}\left[\sum_i x_i\right]$$

$\mathbb{E}[cX] = c \mathbb{E}[X]$, with $c = 1/n$.

$$= \frac{1}{n} \sum_i \mathbb{E}[x_i]$$

$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y]$, applied $n - 1$ times.

$$= \frac{1}{n} \sum_i \mu$$

Identically distributed: every draw has mean μ .

$$= \frac{1}{n} (n\mu) = \mu$$

The sum of n copies of μ .

$\mathbb{E}[\bar{X}] = \mu$: across repeated samples from the same process, the sample means center on the population mean.

The variance of the sample mean

$$\mathbb{V}[\bar{X}] = \mathbb{V}\left[\frac{1}{n} \sum_i X_i\right]$$

The definition of $\bar{X}$.

$$= \frac{1}{n^2} \mathbb{V}\left[\sum_i X_i\right]$$

$\mathbb{V}[cX] = c^2 \mathbb{V}[X]$, with $c = 1/n$.

$$= \frac{1}{n^2} \sum_i \mathbb{V}[X_i]$$

Zero covariance, due to independence.

$$= \frac{1}{n^2} \sum_i \sigma^2$$

Identically distributed: every draw has variance σ^2 .

$$= \frac{1}{n^2} (n\sigma^2) = \frac{\sigma^2}{n}$$

The sum of n copies of σ^2 .

Where the independence entered

$$\mathbb{V}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{V}[X_i] + \underbrace{2 \sum_i \sum_{j>i} \text{Cov}(X_i, X_j)}_{= 0 \text{ when the } X_i \text{ are independent}}$$

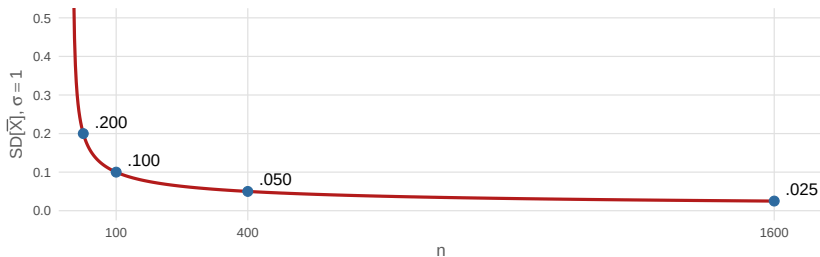
When the X_i are correlated, the covariance terms need not be zero. $\mathbb{V}[\bar{X}]$ exceeds σ^2/n when they sum to a positive number. It falls below σ^2/n when they sum to a negative number. We will return to this later in the semester.

Relationship between n and standard deviation

The standard deviation of an estimator's sampling distribution is its **standard error**:

$$SD[\tilde{X}] = \frac{\sigma}{\sqrt{n}}$$

n	25	100	400	1600
$SD[\tilde{X}], \text{ at } \sigma = 1$	.200	.100	.050	.025



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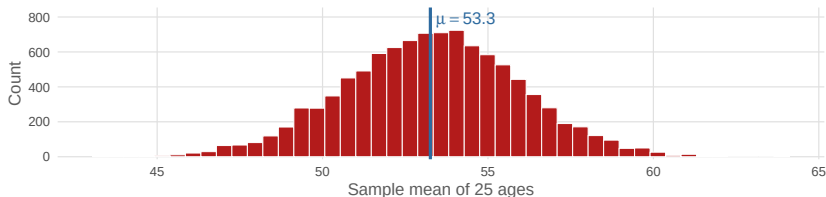
Wrapping up

Sampling distribution

We have established, for any n , $\mathbb{E}[\bar{X}] = \mu$ and $\mathbb{V}[\bar{X}] = \sigma^2/n$.

SAMPLING DISTRIBUTION

The **sampling distribution** of an estimator $\hat{\theta}$ is its probability distribution over repeated samples of size n from the same process.



Recall: an estimator $\hat{\theta} = g(X_1, \dots, X_n)$ is a function of the sample.

Convergence in probability

CONVERGENCE IN PROBABILITY

A sequence of random variables W_n **converges in probability** to a constant c when, for every $\varepsilon > 0$,

$$\mathbb{P}(|W_n - c| > \varepsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Written $W_n \xrightarrow{p} c$, or $\text{plim } W_n = c$.

ε is any positive distance, however small.

$\mathbb{P}(|W_n - c| > \varepsilon)$ is the probability that W_n lands more than ε away from c .

Converging in probability means this probability goes to zero as n grows, for every $\varepsilon > 0$.

$W_n \xrightarrow{p} c$ is read " W_n converges in probability to c ."

plim stands for "probability limit." $\text{plim } W_n = c$ is another way of writing $W_n \xrightarrow{p} c$.

The law of large numbers

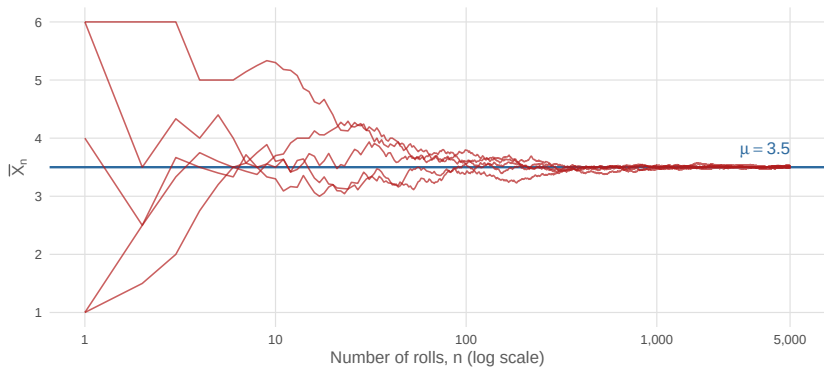
LAW OF LARGE NUMBERS

If $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$ with $\mathbb{E}[X_i] = \mu$ finite, then

$$\bar{X}_n \xrightarrow{p} \mu.$$

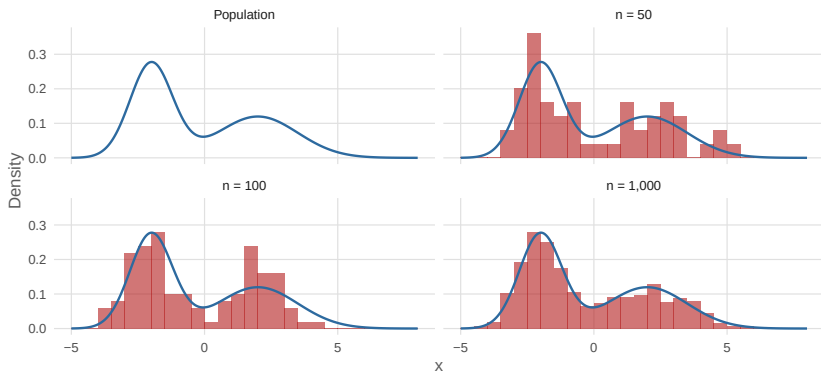
F can be any distribution with a finite mean.

Running means of die rolls



LLN: the distribution of the sample

Another implication is that as the sample size grows large, the distribution of the sample converges to the population distribution.



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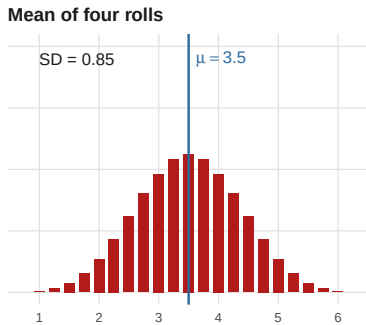
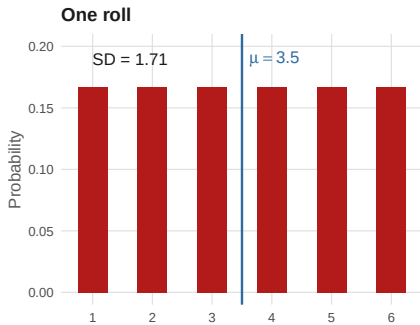
Estimators and their properties

Wrapping up

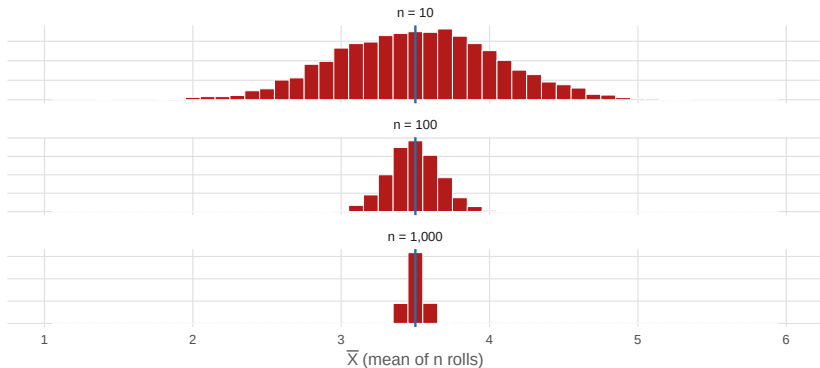
Rolling dice

1. Roll the die four times and add the four results.
2. Write the sum in the log.
3. Mark an $\times$ above that sum on Figure B.
4. Repeat, for 40 trials.

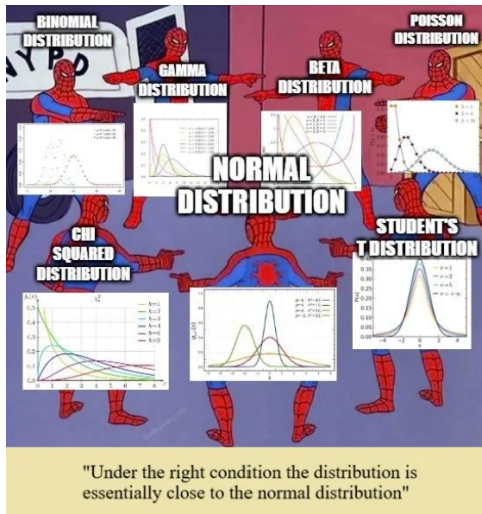
One roll and the mean of four



The mean of n rolls as n grows



The shape of the sampling distribution



The central limit theorem

CENTRAL LIMIT THEOREM

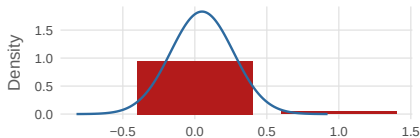
If $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$ with mean μ and finite variance σ^2 , then as $n \rightarrow \infty$

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

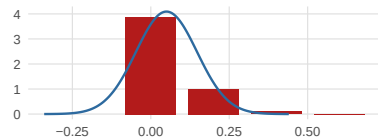
For large n , $\bar{X}$ is approximately $\mathcal{N}(\mu, \sigma^2/n)$, whatever the shape of F . How large n must be depends on F : a skewed F needs a larger n .

A low-turnout election (only 5% of registered voters turn out)

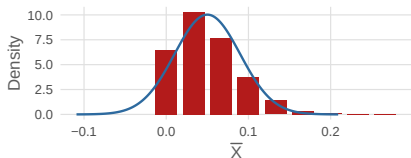
n = 1



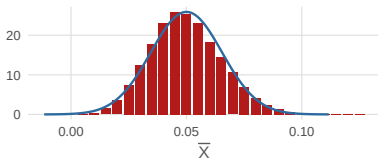
n = 5



n = 30



n = 200



Example: a ballot measure using the CLT vs. simulation

Support for a ballot measure is Bernoulli with $p = .50$. In a poll of $n = 400$ voters, what is $\mathbb{P}(\bar{X} > .55)$?

$$\bar{X} \approx \mathcal{N}\left(.50, \frac{.25}{400}\right), \quad \text{SD}[\bar{X}] = \sqrt{.25/400} = .025$$

```
1 - pnorm(0.55, mean = 0.50, sd = 0.025)
```

```
[1] 0.02275013
```

Using simulation:

```
set.seed(6019)
# 10,000 polls of 400 voters; the mean of each
xbar <- replicate(10000, mean(rbinom(400, 1, 0.5)))
# the share of those means above .55
mean(xbar > 0.55)
```

```
[1] 0.02
```

The gap: $\bar{X}$ moves in steps of $1/400$. The normal curve is continuous.

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Learning from the random sample

I want to know what share of American adults identify as Democrats. Say I take a simple random sample of American adults, and each tells me whether they identify as a Democrat.

Question: Did I answer my question?

Answer: No. But I learned about the answer to my question.

Inference about a parameter

Denote θ as the thing I want to learn about: here, the share of American adults who identify as Democrats.

- θ is an unknown population parameter.
- I can use various procedures to construct estimates of θ , which we denote $\hat{\theta}$ (read this as “theta-hat”).
- Here, we select a sample of size n and calculate the share of Democrats in that sample. With $X_i = 1$ if respondent i identifies as a Democrat and $X_i = 0$ otherwise,

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}.$$

- The sample proportion is the sample mean of a 0/1 variable. It is an estimator of the population proportion.

BIAS

The **bias** of an estimator $\hat{\theta}$ of θ is $\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$.
 $\hat{\theta}$ is **unbiased** when $\text{Bias}(\hat{\theta}) = 0$.

- The sample mean is an **unbiased estimator** of the population mean:
 $\mathbb{E}[\bar{X}] = \mu$.
- Note: this is independent of the sample size.

CONSISTENCY

An estimator $\hat{\theta}_n$ is **consistent** for θ when $\hat{\theta}_n \xrightarrow{P} \theta$.

- The sample mean is a consistent estimator of the population mean because of the law of large numbers.
- The share of Democrats among the first four respondents is an unbiased but inconsistent estimator of θ .
- The sample mean plus a constant is a biased and inconsistent estimator of the population mean.
- $\bar{X} + 1/n$ is a biased but consistent estimator of the population mean.

EFFICIENCY

Of two unbiased estimators of θ , the one with the smaller variance is the more **efficient**.

- The share of Democrats among the first four respondents is an unbiased but inconsistent estimator of θ .
- But the share among the first ten is a more efficient (but still inconsistent) estimator of θ .
- A biased estimator can have a smaller variance than an unbiased one.

NB: Statistical theorists investigate the absolute or asymptotic efficiency of estimators, but that's beyond our scope.

$\bar{X}$ is the most efficient weighted average

Any weighted average of the responses, $\hat{\mu} = \sum_i w_i X_i$ with $\sum_i w_i = 1$. The share among the first four, the share among the first ten, and $\bar{X}$ are all of this form.

$$\mathbb{E}[\hat{\mu}] = \sum_i w_i \mu = \mu$$

Unbiased, whatever the weights.

$$\mathbb{V}[\hat{\mu}] = \sigma^2 \sum_i w_i^2$$

The responses are independent.

$$\sum_i w_i^2 = \frac{1}{n} + \sum_i \left(w_i - \frac{1}{n} \right)^2$$

Smallest when every $w_i = 1/n$.

$$\mathbb{V}[\bar{X}] = \sigma^2/n$$

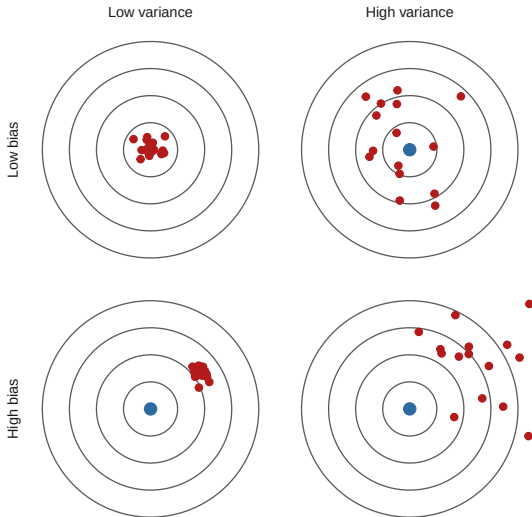
The least variance of any unbiased weighted average.

If bias and efficiency are about the expected value and variance of an estimator, error is about the actual value of a sample estimate compared to a population parameter:

$$\text{Error} = \hat{\theta} - \theta$$

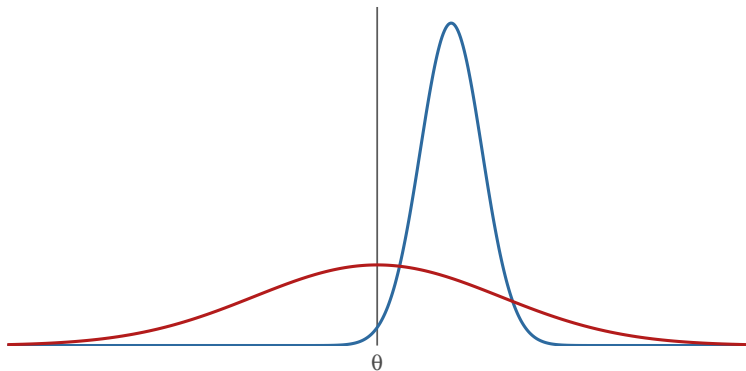
- We cannot normally know the error of an estimate because we don't know θ (that's why we're estimating it).
- Of two unbiased estimators, the more efficient one has smaller squared errors on average.

Target practice



Bias-variance tradeoffs

Would you prefer no bias, or lower variance?



Mean squared error

MEAN SQUARED ERROR

The **mean squared error** of $\hat{\theta}$ is

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] .$$

MSE = Bias² + Variance

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2]$$

The definition.

$$= \mathbb{V}[\hat{\theta} - \theta] + (\mathbb{E}[\hat{\theta} - \theta])^2$$

$\mathbb{E}[W^2] = \mathbb{V}[W] + (\mathbb{E}[W])^2$, with $W = \hat{\theta} - \theta$.

$$= \mathbb{V}[\hat{\theta}] + (\mathbb{E}[\hat{\theta}] - \theta)^2$$

θ is a constant.

$$= \mathbb{V}[\hat{\theta}] + \text{Bias}(\hat{\theta})^2$$

The definition of bias.

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Properties of an estimator

Property	Definition	For $\bar{X}$
Unbiased	$\mathbb{E}[\hat{\theta}] = \theta$	$\mathbb{E}[\bar{X}] = \mu$
Mean squared error	$\mathbb{E}[(\hat{\theta} - \theta)^2] = \mathbb{V} + \text{Bias}^2$	σ^2/n
Efficient	smaller $\mathbb{V}[\hat{\theta}]$ among unbiased	least variance among unbiased weighted averages
Consistent	$\hat{\theta}_n \xrightarrow{P} \theta$	law of large numbers
Asymptotically normal	$\frac{\hat{\theta}_n - \theta}{\text{SD}[\hat{\theta}_n]} \xrightarrow{d} \mathcal{N}(\mathbf{0}, 1)$	central limit theorem

Further reading

- Gailmard, *Statistical Modeling and Inference for Social Science*, ch. 7 and section 9.2.
- Aronow and Miller, *Foundations of Agnostic Statistics*, ch. 3.

These slides draw on teaching materials by Talbot Andrews, Matt Blackwell, Daniel de Kadt, Tom Pepinsky and Teppei Yamamoto.